MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER INSULATED, EXTRUDED TFE, POLYIMIDE COATED, NICKEL-COATED HIGH STRENGTH COPPER ALLOY CONDUCTOR, 600-VOLT

This specification is approved by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759.

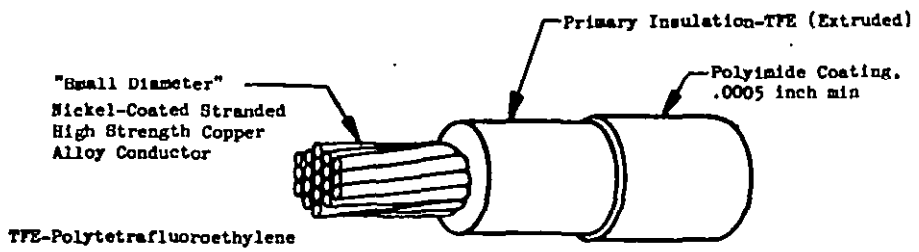


TABLE I. CONSTRUCTION DETAILS

Part No. 1/	Wire size	Stranding (Number of strands X AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire		
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M22759/31-28-*	28	7 X 36	.014	0.016	79.0	.034 ±.002	1.37
M22759/31-26-*	26	19 X 38	.018	0.020	49.4	.039 ±.002	1.99
M22759/31-24-*	24	19 X 36	.023	0.025	30.1	.044 ±.002	2.73
M22759/31-22-*	22	19 X 34	.029	0.031	18.6	.050 ±.002	3.77
M22759/31-20-*	20	19 X 32	.037	0.040	11.4	.059 ±.002	5.55

1/ PART NUMBERS AND AVAILABLE INSULATION COLORS: The asterisks in the part number column, Tables I and II, shall be replaced by numerical color designators in accordance with the following:

Black	0	Orange	3
Brown	1	Yellow	4
Red	2	Green	5

The wires of this specification sheet are presently available in the above-listed solid colors only. Insulation colors blue, violet, gray, and white are not available. See also "COLOR" under "ADDITIONAL REQUIREMENTS", page 2 of this specification sheet.

FSC 6145

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TABLE II. PERFORMANCE DETAILS

Part No.	Abrasion resistance (Procedure II)				Bend testing			
					Mandrel diameter (inches) (±3%)		Test load (lbs) (±3%)	
	Resistance (inches of tape) (min) (initial condition)	Weight support bracket	Weight (lbs)	Tension load (lbs)	Life cycle (oven & bend tests) 1/	Cold bend test	Life cycle (oven & bend tests) 1/	Cold bend test
M22759/31-28-*	15	A	.50	1.0	.125	.250	.50	
M22759/31-26-*	21	A	.50	1.0	.125	.250	.50	
M22759/31-24-*	21	A	.50	1.0	.125	.250	.50	
M22759/31-22-*	27	A	.50	1.0	.250	.375	.75	
M22759/31-20-*	27	A	.50	1.0	.250	.375	.75	

1/ Also for bend tests after immersion.

WIRE RATINGS AND ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 260°C (500°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
ABRASION RESISTANCE AFTER IMMERSION: No requirement
ACID RESISTANCE: No requirement
BLOCKING: 260 ±2°C (500 ±3.6°F)
COLOR: The color limits of MIL-STD-104 are not applicable to the insulation colors of this specification sheet.
DIELECTRIC TEST AFTER IMMERSION: 2200 volts (rms), 60Hz
FLAMMABILITY: Post-flame dielectric test not required
HUMIDITY RESISTANCE: No requirement
IDENTIFICATION OF PRODUCT: White print preferred to provide greater contrast with inherent dark colors of this wire
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
IMPULSE DIELECTRIC TEST: 8.0 kilovolts (peak), 100% test
INSULATION RESISTANCE: 2500 megohms for 1000 ft (min)
LIFE CYCLE: Oven temperature 275 ±2°C (527 ±3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
Note: Wrinkling of polyimide coating in the bend test of the life cycle specimen shall not be cause for rejection.
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
POLYIMIDE CURE TEST: Required. The polyimide coating shall not crack when tested by the procedure shown on page 3 of this specification sheet. For wires of this specification sheet, the polyimide cure test shall be part of the Group II tests of the MIL-W-22759 Quality Conformance Inspection and shall be subject to the inspection level and acceptable quality level specified for the Group II tests.
SHRINKAGE: 0.03 inch max at 290 ±2°C (554 ±3.6°F)
SMOKE: 290°C (554°F)
SPARK TEST OF PRIMARY INSULATION: 1500 volts (rms), 60Hz, 100% test
SURFACE RESISTANCE: 5 megohm-inches (min), initial and final readings
THERMAL SHOCK: Oven temperature, 260 ±2°C (500 ±3.6°F)
Max change in measurement, 0.060 inch
WICKING: No requirement
WIRE LENGTH REQUIREMENTS: Schedule A
WRAP TEST: "Wrap back" test required; no cracking
Oven temperature, 313 ±2°C (595.4 ±3.6°F)

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CURE TEST FOR POLYIMIDE COATING

Two hundred milliliters of distilled water together with a few boiling chips or beads shall be placed in a 1 liter Erlenmeyer flask and the flask shall be closed by a rubber stopper fitted with a water cooled reflux condenser. The flask shall be heated by hot plate or heating mantle until the water is boiling and condensate is returning from the reflux condenser. One end of an approximately 12 inch length of the wire to be tested shall be inserted into the flask by passing it between the rubber stopper and the side of the flask or through a snugly fitting hole in the stopper, so that 5 inches of the wire length extends into the vapor phase inside the flask. The portion of the wire inside the flask shall be essentially straight and shall not be in contact with the glass sides of the flask or condenser, the layer of liquid water in the bottom of the flask, or the liquid condensate returning from the condenser. Heating of the flask shall be resumed, with stopper and reflux condenser again in place. The portion of wire inside the flask shall be exposed to the vapor phase above the boiling water for exactly one hour and shall then be removed from the flask. A 4 inch specimen shall be cut from the vapor-exposed portion of the wire, avoiding the one inch which was nearest the rubber stopper during vapor exposure. The 4 inch specimen shall be allowed to cool at room temperature for a minimum of fifteen minutes, after which it shall be wrapped in a tight spiral for six turns or the full length of the specimen, whichever is lesser, around a mandrel which for wire sizes 18 and smaller shall be the specified maximum diameter of the wire and for wire sizes 16 and larger shall be three times the specified maximum diameter of the wire. The specimen shall then be inspected visually for cracks without the aid of magnification.

Preparing activity:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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